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MILITARY STANDARD

METHOD AND PROCEDURE FOR OVERALL CALIBRATION OF ILLUMINOMETER FOR USE IN ACCEPTANCE TESTING OF PYROTECHNIC ILLUMINANTS



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OFFICE OF THE ASSISTANT SECRETARY OF DEFENSE
WASHINGTON 25, D. C.

20 JULY 1956

Supply and Logistics

Method and Procedures for Overall Calibration
Of Illuminometer for Use in Acceptance Testing
Of Pyrotechnic Illuminants

MIL–STD–191A

1. This standard has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force, effective immediately.

2. In accordance with established procedure, the Standardization Division has designated the Ordnance Corps, and Bureau of Ordnance, respectively, as Army-Navy custodians of this standard.

3. Recommended corrections, additions, or deletions should be addressed to the Standardization Division, Office of the Assistant Secretary of Defense (Supply and Logistics), Washington 25, D. C.

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CONTENTS

1. SCOPE
2. REFERENCED DOCUMENTS
3. DEFINITIONS
4. GENERAL REQUIREMENTS OR STATEMENTS
5. DETAILED REQUIREMENTS OR STATEMENTS
5.1 Description of Test
5.2 Criteria for Passing Test
5.3 Equipment
5.4 Test Procedure
5.5 Method of Calibrating Secondary Standard Working Lamps
10. APPENDIX
10.1 References
10.2 General Information

FIGURES

Figure
1 Detailed Layout of Sector Disk
2 Photometric Arrangement

TABLE

Table I. Example of an Illuminometer Calibration Report

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1. SCOPE

1.1 SCOPE. The purpose of this standard is to establish a method and procedure for overall calibration of illuminometers used in acceptance testing of pyrotechnic illuminants. This procedure is applicable to illuminometers used to determine candlepower and color value of pyrotechnic flares and signals.

2. REFERENCED DOCUMENTS

2.1 The following specifications, of the issue in effect on date of invitation for bids, form a part of this standard:

MILITARY SPECIFICATIONS

MIL–P–20464 — Pyrotechnics: Method of Determination of Candlepower and Color Value.

(Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. DEFINITIONS

3.1 C.I.E. The international accepted abbreviation for the International Conference on Illumination (French-Commission Internationale De L'Eclairage).

3.2 ILLUMINOMETER. An instrument for measuring incident light, consisting of a microameter (hereinafter referred to as the meter) calibrated in foot-candles, and a barrier layer cell pick-up (hereinafter referred to as the target) corrected to a C.I.E. response by means of a viscor filter.

3.3 CARDINAL POINT. A definite, marked division on the scale of the meter.

3.4 STANDARD LAMP. A tungsten filament lamp (hereinafter referred to as primary standard lamp) selected and calibrated by the National Bureau of Standards (see 5.3.1).

3.5 WORKING STANDARD LAMP. A tungsten filament lamp (hereinafter referred to as a secondary standard lamp) selected and calibrated against a primary standard lamp (see 5.5).

4. GENERAL REQUIREMENTS OR STATEMENTS NOT APPLICABLE

5. DETAILED REQUIREMENTS OR STATEMENTS

5.1 DESCRIPTION OF TEST.

5.1.1 The test consists of checking the accuracy of viscor filters provided with the illuminometer to corect the barrier layer cell to the C.I.E. "Y" response which approximates that of the human eye; and determining a correction factor to be used for intensity (candlepower) measurements at several points on each scale of the meter (see table I).

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TABLE I. *Example of an Illuminometer Calibration Report*

1. *Illuminometer tested:* Weston Model No. 603, Serial No. 2173
   Date of test: 1/1/53
   Candlepower of lamp at 2854°K color temperature: 1455
   Tested by: J. Doe

2. *Color response*

Filter designation	*Percent luminous transmission of filters employing a standard lamp of 2854°K color temperature*	
	Values obtained with illuminometer being tested	*Values obtained at the NBS*
Red, Corning No. 2412	13.5	13.7
Green, Corning No. 4407	24.5	23.5

3. *Candlepower correction factors:*

Meter scale	*Meter read. foot-candle*	*Sector disk transmission*	*Test distance feet*	*Incident light foot-candles*	*Correction factor*
0–10	2.5	10%	7.79	2.4	0.96
0–10	5.0	20%	7.79	4.8	.95
0–10	7.5	30%	7.79	7.1	.94
0–50	10	30%	6.74	9.6	.96
0–50	25	—	7.82	23.8	.95
0–50	40	—	6.22	37.6	.94
0–500	100	—	3.91	95.0	.95
0–500	250	—	2.49	235	.94
0–500	450	—	1.86	419	.93

4. *Remarks:*

The Illuminometer tested is acceptable.

5.2 CRITERIA FOR PASSING TEST.

5.2.1 The illuminometer shall comply with the overall calibration requirements as specified in this standard.

5.3 EQUIPMENT.

5.3.1 *Standard lamps.* Airway beacon or projector lamps having tungsten filaments, and ratings of 500, 1000, and 1500 watts, are suitable for use as light sources for checking illuminometers. These lamps must be calibrated by the National Bureau of Standards (NBS), to determine the voltage and current ratings necessary to obtain a color temperature of 2854°K. The candlepower of these lamps, at this color temperature, must also be certified (see 10.1.1). Working standard lamps shall be obtained by calibrating against an NBS standardized lamp as specified in 5.5.

5.3.2 *Glass filters.* Two glass filters, large enough to cover the illuminometer target, shall be required. The filters shall be a Corning Red Filter No. 2412, and a Corning Green Filter No. 4407, or approved substantial equals. These filters must be calibrated by NBS for percent luminous transmission at a temperature of 25°C, when used with an illuminant having a color temperature of 2854°K.

5.3.3 *Motor.* A fan type motor, 1550 revolutions per minute with a threaded shaft and lock nuts is required.

5.3.4 *Sector disks.* Five sector disks which shall fit the motor shaft and allow 50, 60, 70,

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75 and 80 percent of the light to pass (see figure 1). Lower percentages of light passage may be obtained by combinations of these disks. For example: 50 percent, plus a 60 percent disk, will permit 10 percent of the light to pass.

5.3.5 *Voltage regulation.* Direct current (d.c.) line voltage if available shall be used. If alternating current (a.c.) line voltage is used, it shall not vary more than ± 0.1 percent. A voltage regulator (see 10.2.1) shall be used. The regulator shall adhere to the following requirements:

Input Voltage:	**100 to 130 volts a.c., single phase, 55 to 65 cycles per second.**
Output voltage:	**115 volts a.c.**
Regulation accuracy:	**±1 percent for a variation in input voltage of ±10 percent and ±1 percent for a load change of 100 percent.**

5.3.6 *Rheostat.* A ten ampere, 0–130 volts a.c. or d.c., fine adjusting rheostat.

5.3.7 *Ammeter.* A zero to 10 ampere a.c. or d.c. ammeter, accurate to within ± 0.1 percent (see 5.3.6).

5.3.8 *Voltmeter.* A zero to 120 volts a.c. or d.c. voltmeter, accurate to within ± 0.1 percent (see 5.5.3.7).

5.3.9 *Optical bench.* A Gaertner, 3 meters or longer, or approved substantial equal.

5.3.10 *Lamp sockets.* Lamp sockets to accommodate the primary standard and secondary standard lamp shall be mounted on the optical bench.

5.3.11 *Black velvet and photometric screening.* Black velvet and photometric screening shall be mounted on the optical bench to eliminate all external and reflected light.

5.4 TEST PROCEDURE.

5.4.1 *General.* A secondary standard lamp, with a color temperature of 2854°K, shall be set up at one end of the optical bench (see figure 2). The voltmeter and ammeter shall be connected, plugged into the rheostat and the rheostat connected to a regulated power supply of 115 volts a.c. or d.c. The illuminometer target shall be mounted on the bench. The photometric screening and black velvet shall be arranged so that only the light from the standard lamp will reach the cells. The diaphragm dimensions, specified by NBS on the lamp certificate, shall be used to limit the light to that coming from the filament only, and not from the reflecting surface of the lamp bulb. The lamp and the target shall be oriented so that the face of the cells and the filament of the lamp will be at the same height; in the same plane; and perpendicular to the optical axis, as determined by the two lines etched on the opposite sides of the lamp. The etched line with the circle shall be turned away from the target.

5.4.1.1 The current shall be adjusted so that the lamp will be operated within ± 0.25 percent of its rated voltage and current. The voltmeter shall be removed from the circuit when the current is read. The voltage drop, across the ammeter, shall be computed, and the voltmeter reading corrected accordingly.

5.4.1.2 If the simultaneous corrected readings of the voltmeter and ammeter do not check with the values reported by NBS for the lamp under test, the lamp shall be replaced with another. If the readings still do not check, either the voltmeter or the ammeter or both are in error, and shall be recalibrated before proceeding with the test.

5.4.2 *Accuracy check (color response) of the viscor filter supplied with the illuminometer.*

5.4.2.1 The distance, from the lamp to the illuminometer target, shall be adjusted until the meter indicates exactly 100 foot-candles.

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5.4.2.2 A calibrated rotating sector disk, with a transmission of exactly 20 percent, shall be placed between the lamp and the illuminometer target, and the reading of the illuminometer shall be noted. A correction factor, for the reading, shall be calculated as follows:

$$F = \frac{100 \times 0.20}{R}$$

where:

F = Correction factor for the illuminometer meter reading.

R = Reading of the illuminometer meter.

The sector disk shall be removed and the illuminometer meter reading checked to insure that it still reads exactly 100 foot-candles. In lieu of a sector disk, if the length of the optical bench permits, the distance from lamp to target needed to give 100 foot-candles may be doubled, thus giving 25 foot-candles.

5.4.2.3 A green glass filter, Corning No. 4407, or approved substantial equal, calibrated by NBS, shall be installed in place of the sector disk, and the illuminometer meter reading shall be noted. This filter shall be used at 25°C. ± 2°C. The percent of luminous transmission shall be calculated as follows:

$$Tg = Rg \times F$$

where:

Tg = Percent of luminous transmission.

Rg = Illuminometer meter reading.

F = Correction factor for the illuminometer meter reading.

5.4.2.4 The procedure in 5.4.2.3 shall be repeated using a red glass filter, Corning No. 2412, or approved substantial equal, calibrated by NBS. If the luminous transmission, as determined with the illuminometer, differs from that determined by NBS by more than 5 percent, the illuminometer shall be returned to the manufacturer with a recommendation that it be repaired calling attention to the fact that this is an illuminometer used in acceptance testing of pyrotechnic illuminants for the Government. For example, if the NBS certified luminous transmission for a red filter is 13.7 percent, the maximum permissible deviation as measured by the illuminometer is (13.7 × 0.05) or ± 0.7 percent. Acceptable transmission would therefore be from 13 to 14.4 percent.

5.4.3 *Accuracy check of the illuminometer for intensity measurements.*

5.4.3.1 If the color response of the illuminometer target is satisfactory, the illuminometer meter shall then be checked for accuracy of intensity measurements, and suitable correction factors determined.

5.4.3.2 The lamp and illuminometer shall be set up to comply with 5.4.1. The distance between the lamp and the illuminometer target shall be varied until the illuminometer meter indicates a cardinal point. The distance shall then be measured, from the face of the illuminometer target to the center of the lamp filament, to an accuracy of 0.005 feet (0.06 inches). Such readings shall be taken at three points, on each scale of the meter, at approximately 25, 50, and 75 percent of full scale. For example, a Weston Model 756 Illuminometer having ranges of 0 to 6, 0 to 12, 0 to 60, 0 to 120, and 0 to 600 foot-candles, should be checked at 1.5, 3, and 4.5 for the 0 to 6 range; 2, 6, and 10 for the 0 to 12 range, etc. If the optical bench is not long enough to permit readings on the lower scales, a lower candlepower lamp or the calibrated sector disks may be used.

5.4.3.3 A check of the photometric screening shall be made each time the distance is changed, to insure that the screening is effective in preventing reflected light from reaching the target.

5.4.3.4 The illuminometer readings, when multiplied by the square of the distance, give

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the indicated candlepower of the lamp. The factor, used as a multiplier to correct the illuminometer meter reading, is calculated as follows:

F_c = Rated candlepower of lamp divided by the indicated candlepower of lamp,

where:

F_c = Illuminometer meter reading correction-factor.

5.4.3.5 If these factors are found to be greater than 1.1 or less than 0.9, the meter shall be returned to the manufacturer with a recommendation that it be repaired in order to comply with Specification MIL–P–20464.

5.4.4 *Check of the period of response.* The distance between the lamp and the illuminometer shall be adjusted until the meter reads near full scale. The time interval between the instant of exposure of the illuminometer to the light source and the instant at which 90 percent of the peak reading of the illuminometer is reached is then measured. This shall be repeated for each scale on the meter and shall not exceed 3 seconds.

5.5 METHOD OF CALIBRATING SECONDARY WORKING STANDARD LAMPS.

5.5.1 *Description of Test.* This test procedure consists of a comparator method to determine satisfactory values of color temperature and luminous intensity for the lamps to be employed as secondary working standard lamps.

5.5.2 *Criteria for passing test.* The lamps selected to be employed as secondary working standards shall comply with the overall calibration requirements as specified within this test procedure.

5.5.3 *Equipment.*

5.5.3.1 *Standard lamps.* In order to establish a secondary or working standard lamp, it is necessary to have at least three lamps of approximately the same wattage rating which have been standardized by NBS for color temperature (2854°K) and luminous intensity (see 5.3.1).

5.5.3.2 *Voltage regulation.* See 5.3.5.

5.5.3.3 *Optical bench.* See 5.3.9.

5.5.3.4 *Illuminometer.* Weston Model 756, or approved substantial equal.

5.5.3.5 *Rheostat.* See 5.3.6.

5.5.3.6 *Ammeter.* Weston Model 326, or approved substantial equal.

5.5.3.7 *Voltmeter.* Weston Model 326, or approved substantial equal.

5.5.3.8 *Photometric screening.* See 5.3.11.

5.5.4 *Procedure.*

5.5.4.1 The lamps to be standardized shall be carefully selected to be certain that there are no flaws in the glass envelope and that the filament is straight.

5.5.4.2 The lamps shall be aged by burning them at their rated voltage for 2 percent of their rated life. (The rated life shall be obtained from the manufacturer's catalog.) For example, a General Electric 1,000 watt, Type T-20, mogul bipost, aviation lamp, has an average life of 500 hours; therefore, it shall be burned for 10 hours before attempting to standardize it.

5.5.4.3 Two lines shall be etched on opposite sides of the lamp, in a plane at right angles to the plane of the filament. One of these lines shall be marked with a circle to designate the side of the lamp to be placed away from the target when measuring luminous intensity. These etched lines shall be used to insure that the filament will be in the same relative position to the target each time the lamp is used. The lamp shall be sighted through, and the two etched lines shall be in line with the

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target, with the filament at the same height as the target.

5.5.4.4 Before determining the luminous intensity, the lamp shall be adjusted to a color temperature of 2,854°K. This may be done by the method described by King, Snider, and Hamburger (see 10.1.2) or it may be done visually by using the Leeds and Northrup Lummer-Brodhun photometer, comparing it with an NBS standard lamp. It is advisable to use three different NBS standardized lamps in this determination. The current and voltage required by each lamp to obtain the proper color temperature shall be noted. A method of matching color temperature has been developed by Picatinny Arsenal (see 10.13).

5.5.4.5 The luminous intensity can be determined after the requirements in 5.5.4.4 have been complied with. The luminous intensity determinations shall be made on an optical bench, using three primary (NBS standardized) standard lamps and a satisfactory illuminometer. One primary lamp shall be placed at the end of the bench, and the illuminometer shall be placed at a suitable distance so that the illuminometer meter will indicate a cardinal point. The photometric screening shall be used so that only light from the lamp strikes the target. This is very important in order to eliminate errors from reflected light. In addition, all of the light from the filament must strike the target. In order to check this, the eye shall be placed at each of the four corners of the target to make certain that the entire filament is visible from each corner of the target.

5.5.4.6 When the requirements in 5.5.4.5 have been complied with, the primary lamp shall be removed and one of the unknown lamps shall be oriented in its place. The distance from the lamp to the target shall be varied until the same cardinal point obtained in 5.5.4.5 is indicated on the illuminometer meter. It shall be made certain that the photometric screening is adequate. The luminous intensity shall then be computed by means of the inverse square law. Three primary lamps shall be used, and the determinations shall be repeated independently within 24 hours, by another technician as a check against any possible errors. The voltage and current settings shall be checked closely, since a 1 percent change in voltage will cause a 5 percent change in the candlepower of the lamp. Should any of the candlepower values calculated for the secondary lamp being calibrated, based upon the rated luminous intensity values of the three primary lamps, exceed ± 1.0 percent of the mean value of the calculated candlepowers, then either the calibration was faulty or the primary standard lamps are no longer reliable and shall be replaced. For example, the calculated candlepower values of 1500, 1507, and 1520 are obtained for a secondary lamp. The permissible deviations is ± 1.0 percent (or 1507 by 0.01). This indicates that a 15-candle variation about the mean (1,507 candles) exists. It is therefore apparent that this calibration is within the permissible experimental error, and that the calibrations of the primary lamps are still correct.

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Copies of this standard for military use may be obtained as indicated in the foreword to the Index of Military Specifications and Standards.

Copies of this standard may be obtained for other than official use by individuals, firms, and contractors from the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

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APPENDIX

10.1 REFERENCES.

10.1.1 *Test for Schedules of the National Bureau of Standards:*

Test 202.101b—Standardization of seasoned lamps, determination of horizontal candles in a specified direction.

Test 202.107b—Color temperature of incandescent lamps, voltage for a specified color temperature, current for neighboring specified voltage to check permanence.

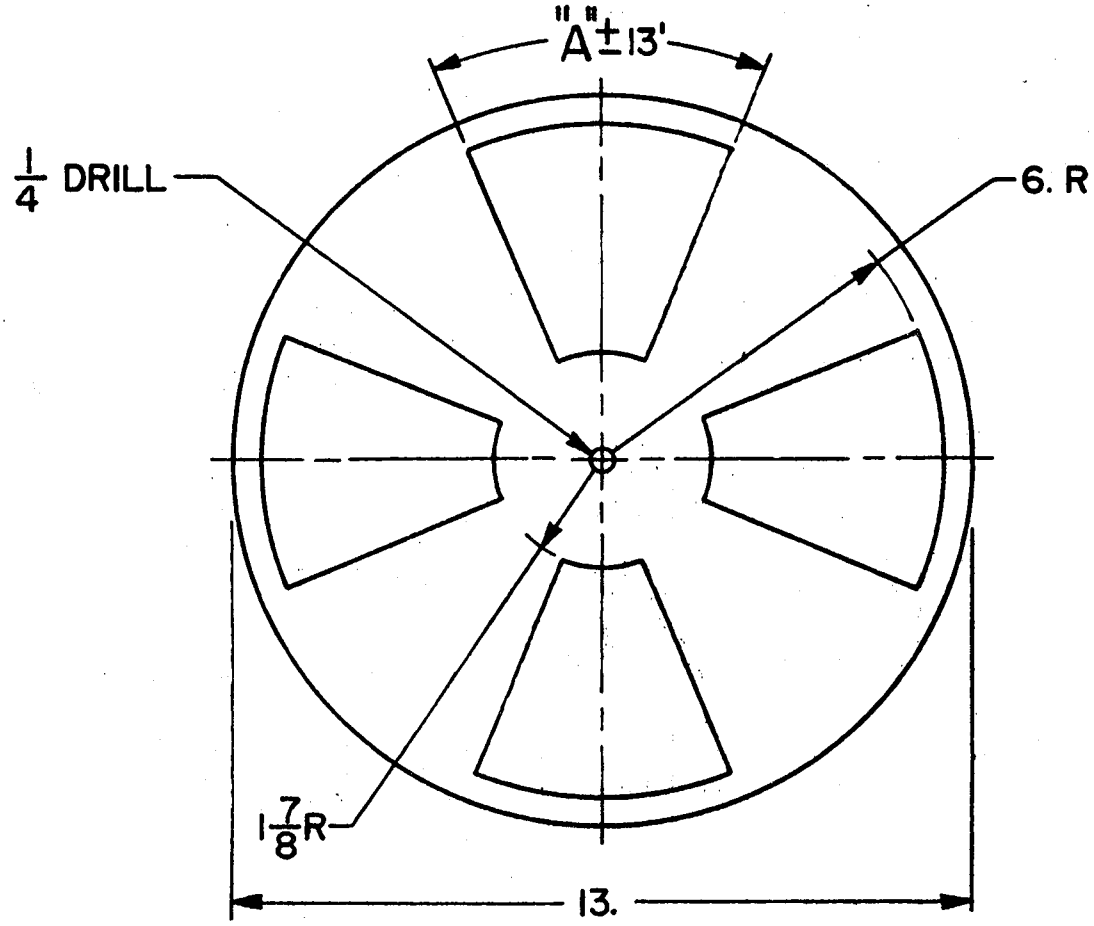
10.1.2 "A Physical Comparator for Color Temperature of Incandescent Lamps," by King, Snider, and Hamburger, Journal of the Optical Society of America, Volume 42, No. 3, page 178, describes a procedure which may be used for adjusting the color temperature of lamps to be used as secondary standards.

10.1.3 Picatinny Arsenal Report, "Instrumentation and Procedure for Establishing Secondary Tungsten Lamps," by Davidson, et al, describes a method of matching color temperature.

10.2 GENERAL INFORMATION.

10.2.1 *Voltage regulator.* The Sorenson Voltage Regulator, Model 1001, has been found to be satisfactory.

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DISC, SECTOR EP–72553 (SERIES C)

1/16 ALUMINUM

1–REQUIRED

"A"	% OPENINGS
45°00'	50
54°00'	60
63°00'	70
72°00'	80
67°30'	75

FIGURE 1. *Detailed layout of sector disc.*

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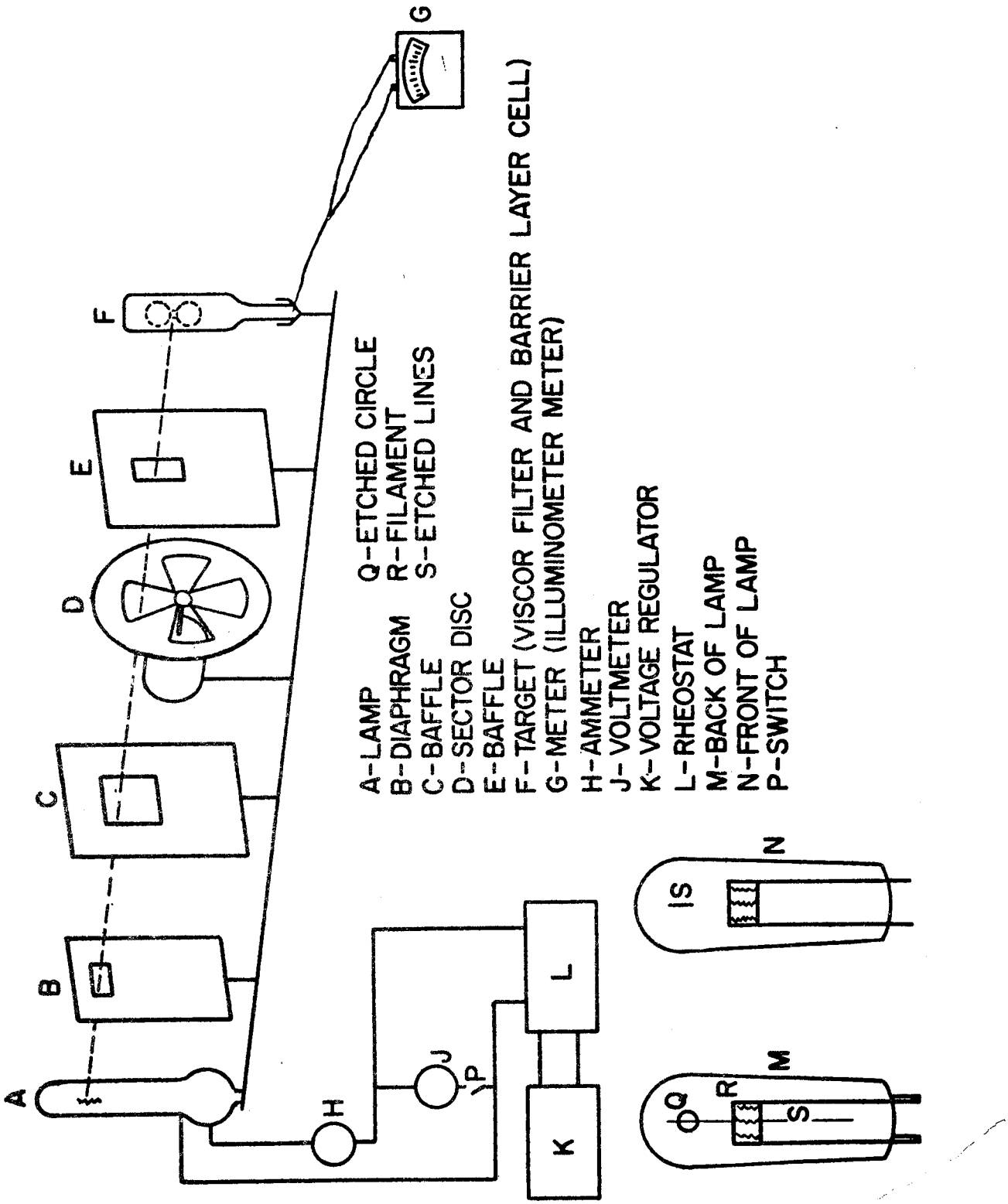


FIGURE 2. *Photometric arrangement.*

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